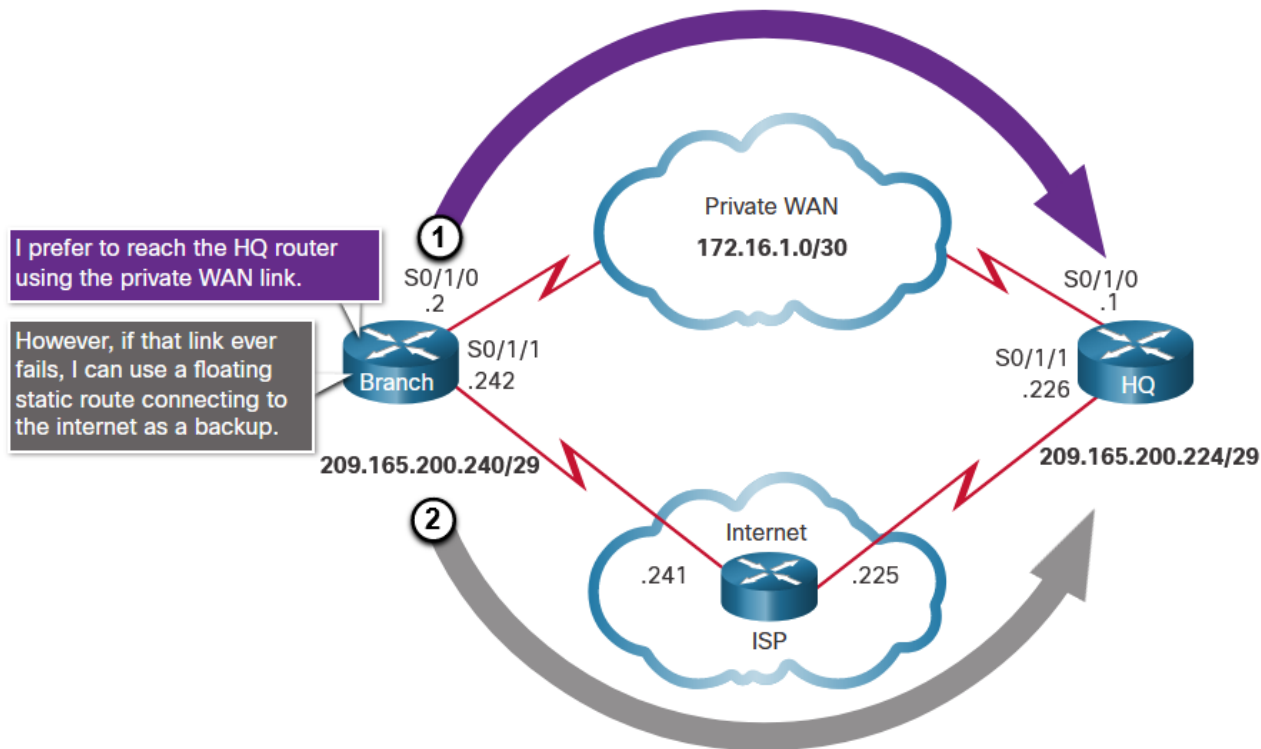


15.4 Configure Floating Static Routes

Configure Floating Static Routes

- Floating static routes are static routes that are used to provide a backup path to a primary static or dynamic route.
- The floating static route is only used when the primary route is not available.



Configure Floating Static Routes

Floating Static Routes

- The floating static route is configured with a **higher administrative distance** than the primary route.
- The administrative distance represents the trustworthiness of a route.
- If multiple paths to the destination exist, the router will choose the path with the lowest administrative distance.

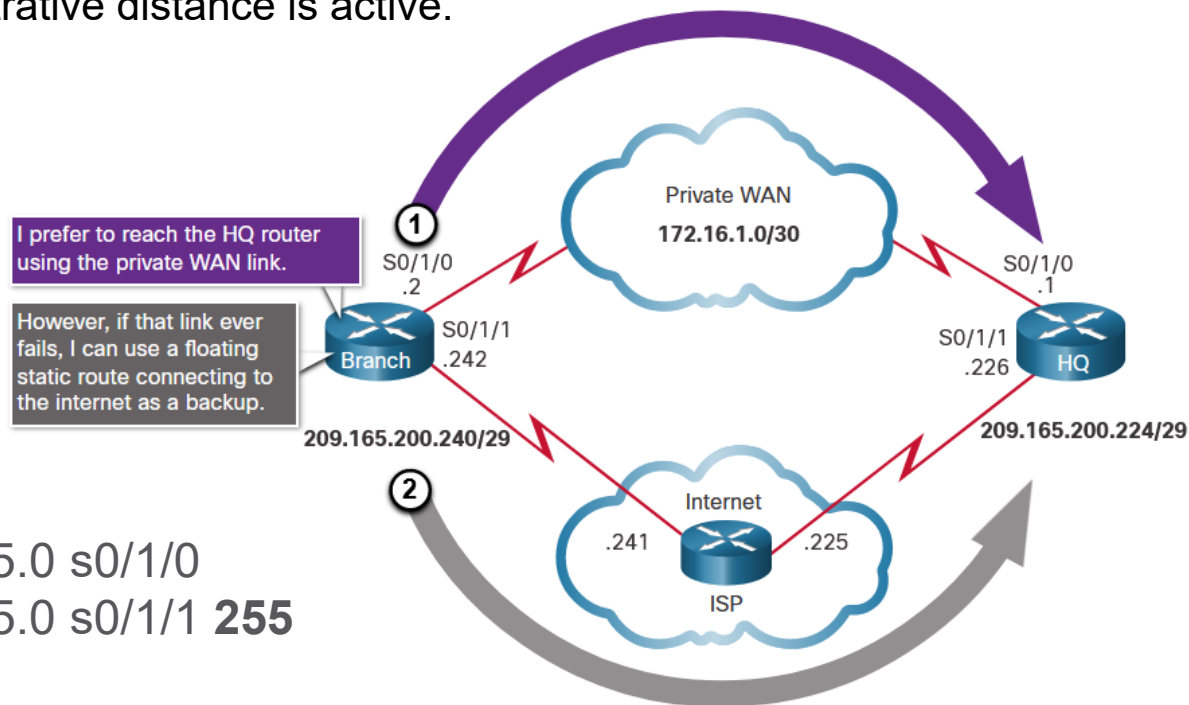
- ## Administrative Distance

- EIGRP = 90
- OSPF = 110
- IS-IS = 115
- Static route = 1

Configure Floating Static Routes

Floating Static Routes

- The administrative distance of a static route can be increased to make the route less desirable than that of another static route or a route learned through a dynamic routing protocol. In this way, the static route “floats” and is not used when the route with the better administrative distance is active.



```
ip route 1.1.1.0 255.255.255.0 s0/1/0
ip route 1.1.1.0 255.255.255.0 s0/1/1 255
```

Configure IPv4 and IPv6 Floating Static Routes

The commands to configure default and floating IP default routes are as follows:

```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.2.2
R1(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.2 5
R1(config)# ipv6 route ::/0 2001:db8:acad:2::2
R1(config)# ipv6 route ::/0 2001:db8:feed:10::2 5
```

The **show ip route** and **show ipv6 route** output verifies that the default routes to R2 are installed in the routing table. Note that the IPv4 floating static route to R3 is not present in the routing table.

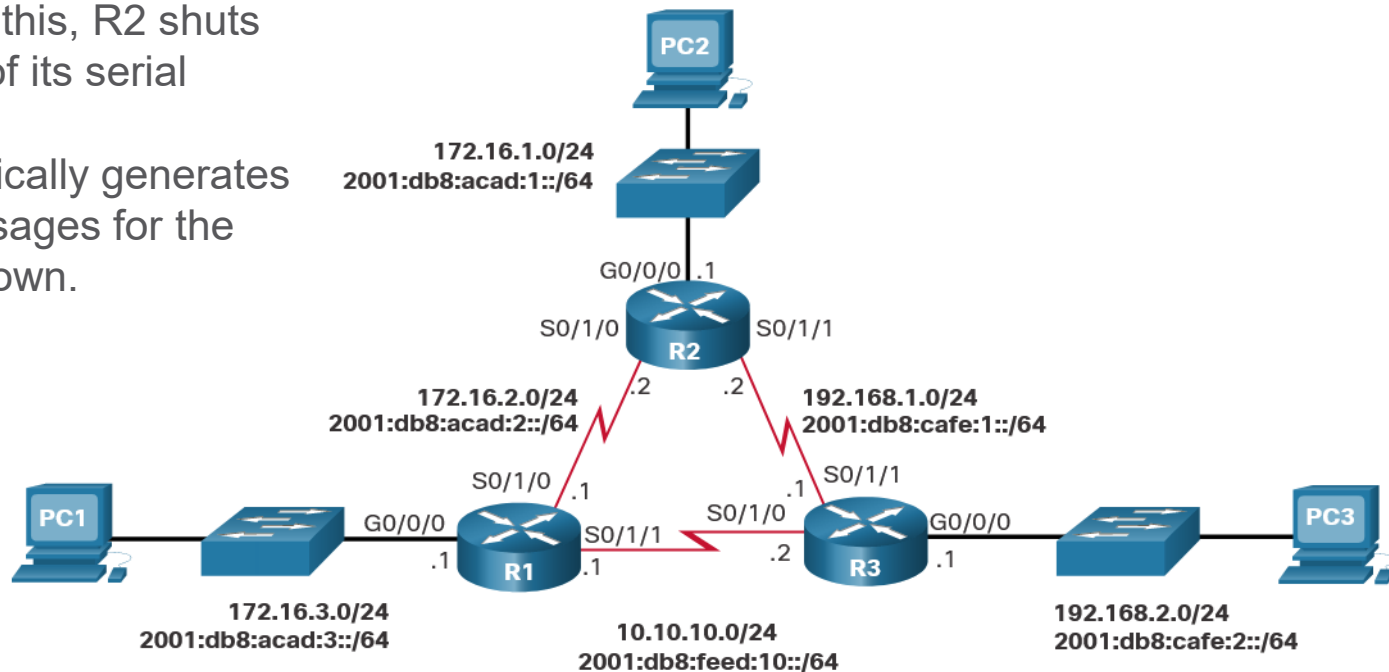
```
R1# show ip route static | begin Gateway
Gateway of last resort is 172.16.2.2 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 172.16.2.2
R1# show ipv6 route static | begin S :
S    ::/0 [1/0]
      via 2001:DB8:ACAD:2::2
R1#
```

Configure Floating Static Routes

Test the Floating Static Routes

- What would happen if R2 failed?
- To simulate this, R2 shuts down both of its serial interfaces.
- R1 automatically generates syslog messages for the link going down.



Configure Floating Static Routes

Test the Floating Static Routes

- A look at R1's routing table would show the secondary route being used.

```
R1# show ip route static | begin Gateway
Gateway of last resort is 10.10.10.2 to network 0.0.0.0
S*    0.0.0.0/0 [5/0] via 10.10.10.2
R1# show ipv6 route static | begin ::
S     ::/0 [5/0]
      via 2001:DB8:FEED:10::2
R1#
```